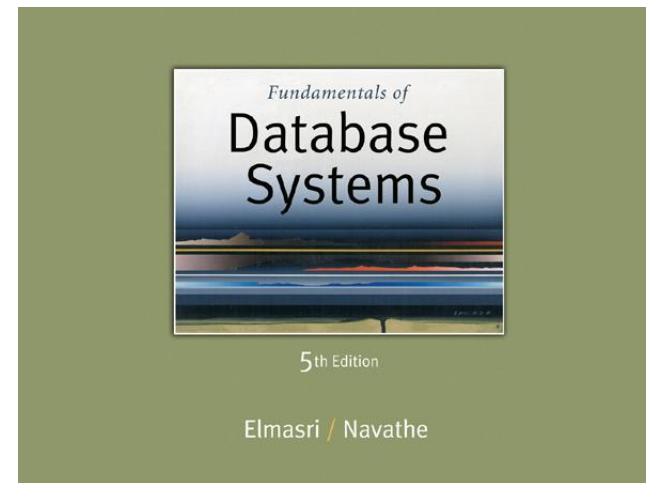


Computer Engineering Program, MCU

CSE 112

Database Management System

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Chapter 1

Introduction: Databases and Database Users



5th Edition

Elmasri / Navathe

Outline

- Types of Databases and Database Applications
- Basic Definitions
- Typical DBMS Functionality
- Example of a Database (UNIVERSITY)
- Main Characteristics of the Database Approach
- Database Users
- Advantages of Using the Database Approach
- When Not to Use Databases

Types of Databases and Database Applications

- Traditional Applications:
 - Numeric and Textual Databases
- More Recent Applications:
 - Multimedia Databases
 - Geographic Information Systems (GIS)
 - Data Warehouses
 - Real-time and Active Databases
 - Many other applications
- First part of book focuses on traditional applications
- *A number of recent applications are described later in the book (for example, Chapters 24,26,28,29,30)*

Traditional applications

- Store structured data: numbers, strings.
- Examples:
 - Student records
 - Banking systems
 - Inventory systems
- Typically implemented using relational databases (MySQL, Oracle, SQL Server).

More Recent Applications

- These handle complex, large-scale, or time-sensitive data:
- Multimedia Databases: Store non-text data
 - Images, audio, video, graphics
 - Example: YouTube, medical imaging systems.
- Data Warehouses: Used for analytics and decision making
 - Historical data
 - Example: Business intelligence systems
- Geographic Information Systems (GIS): Store spatial data:
 - Maps, coordinates, locations
 - Example: Google Maps

What does data warehousing means?

- Collecting data from many different sources and storing it in one central place for analysis and decision making, not for daily operations.
- used by managers and analysts to answer questions such as:
 - What were last year's sales?
 - Which product is most profitable?
 - How did customer behavior change?

Basic Definitions

- **Database:**
 - A collection of related data.
- **Data:**
 - Known facts that can be recorded and have an implicit meaning.
- **Mini-world:**
 - Some part of the real world about which data is stored in a database. For example, student grades and transcripts at a university.
- **Database Management System (DBMS):**
 - A software package/ system to facilitate the creation and maintenance of a computerized database.

Simplified database system environment

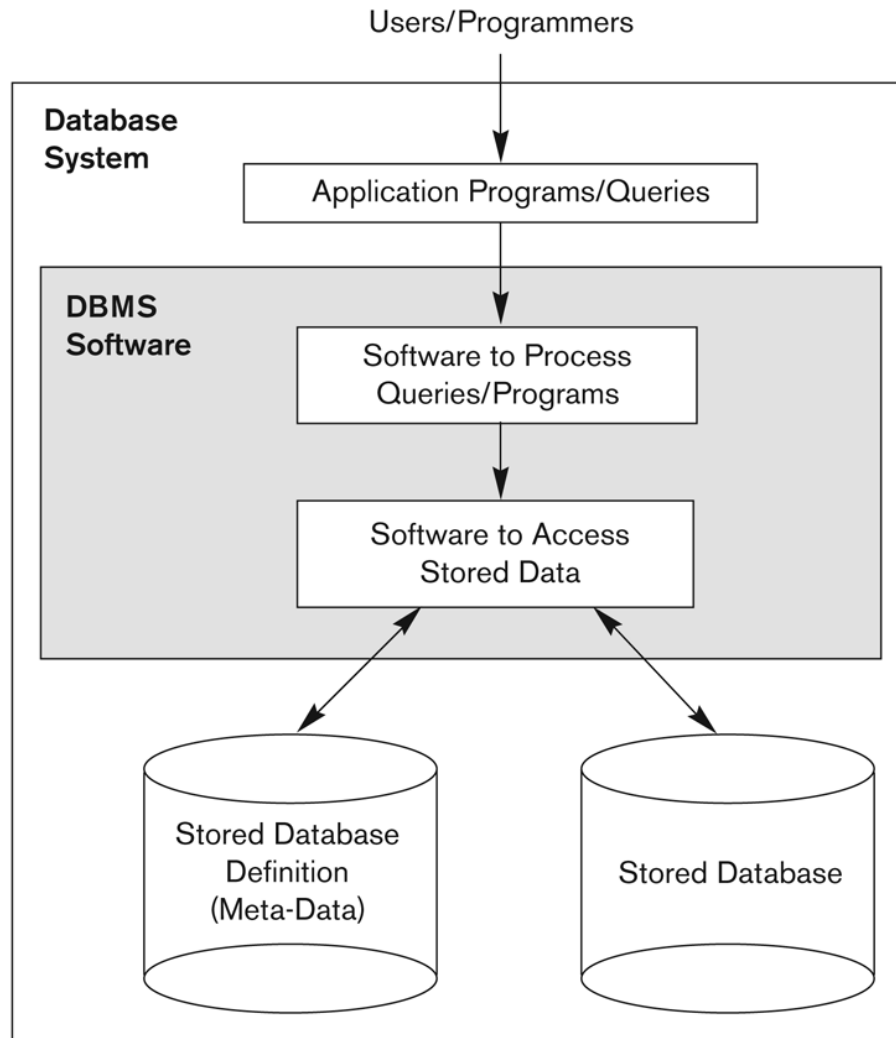


Figure 1.1
A simplified database system environment.

Simplified database system environment

- **Users / Programmers:**
 - End users (students, employees, admins).
 - Or programmers writing application
- **Application Programs / Queries:**
 - This layer is the interface between users and DBMS.
- **DBMS Software (The Brain):**
 - This is the core of the system (MySQL, Oracle, PostgreSQL).
- **Stored Database Definition (Meta-Data): This is data about data:**
 - Table names.
 - Column names
 - Data types
 - Constraints
- **Stored Database:**
 - The actual real data (Students records, orders, employees)

Typical DBMS Functionality

- *Define* a particular database in terms of its data types, structures, and constraints
- *Construct* or Load the initial database contents on a secondary storage medium.
- *Manipulating* the database:
 - Retrieval: Querying, generating reports
 - Modification: Insertions, deletions and updates to its content
 - Accessing the database through Web applications
- Protection or Security measures to prevent unauthorized access
- Presentation and Visualization of data

Example of a Database (with a Conceptual Data Model)

- **Mini-world for the example:**
 - Part of a UNIVERSITY environment.
- **Some mini-world *entities*:**
 - STUDENTs
 - COURSEs
 - Majors (of COURSEs)
 - (academic) DEPARTMENTs
 - INSTRUCTORs

Example of a Database (with a Conceptual Data Model)

- **Some mini-world *relationships*:**
 - Majors *are of specific* COURSEs
 - STUDENTs *take* Courses
 - COURSEs *have prerequisite* COURSEs
 - INSTRUCTORs *teach* COURSEs
 - COURSEs *are offered by* DEPARTMENTs
 - STUDENTs *major in* DEPARTMENTs
- **Note:** The above entities and relationships are typically expressed in a conceptual data model, such as the ENTITY-RELATIONSHIP data model (see Chapters 3, 4)

Example of a simple database

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	04	King
92	CS1310	Fall	04	Anderson
102	CS3320	Spring	05	Knuth
112	MATH2410	Fall	05	Chang
119	CS1310	Fall	05	Anderson
135	CS3380	Fall	05	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

Figure 1.2

A database that stores student and course information.

Example of file processing: Student records in a text file

File: `students.txt`

```
101, Ahmed, 3.5  
102, Sara, 3.9  
103, Omar, 2.8
```

Program (Python):

```
python  
  
file = open("students.txt", "r")  
  
for line in file:  
    data = line.split(",")  
    print(data[1], "GPA =", data[2])  
  
file.close()
```

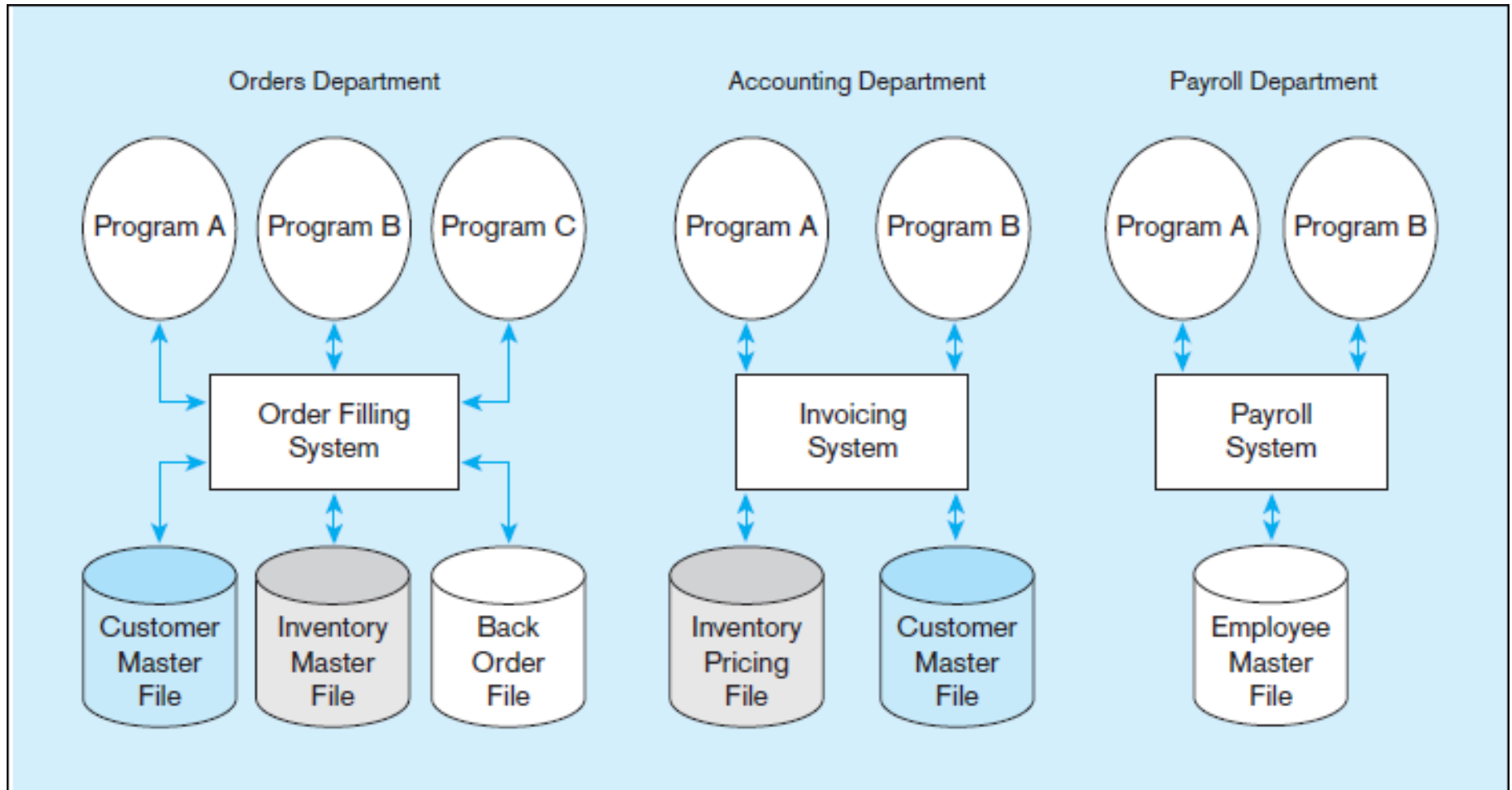
Disadvantages of file processing

- **Data redundancy:** Data redundancy refers to the duplication of data, lets say we are managing the data of a college where a student is enrolled for two courses, the same student details in such case will be stored twice, which will take more storage than needed. Data redundancy often leads to higher storage costs and poor access time.
- **Data inconsistency:** Data redundancy leads to data inconsistency, lets take the same example that we have taken above, a student is enrolled for two courses and we have student address stored twice, now lets say student requests to change his address, **if the address is changed** at one place and not on all the records then this can lead to data inconsistency.

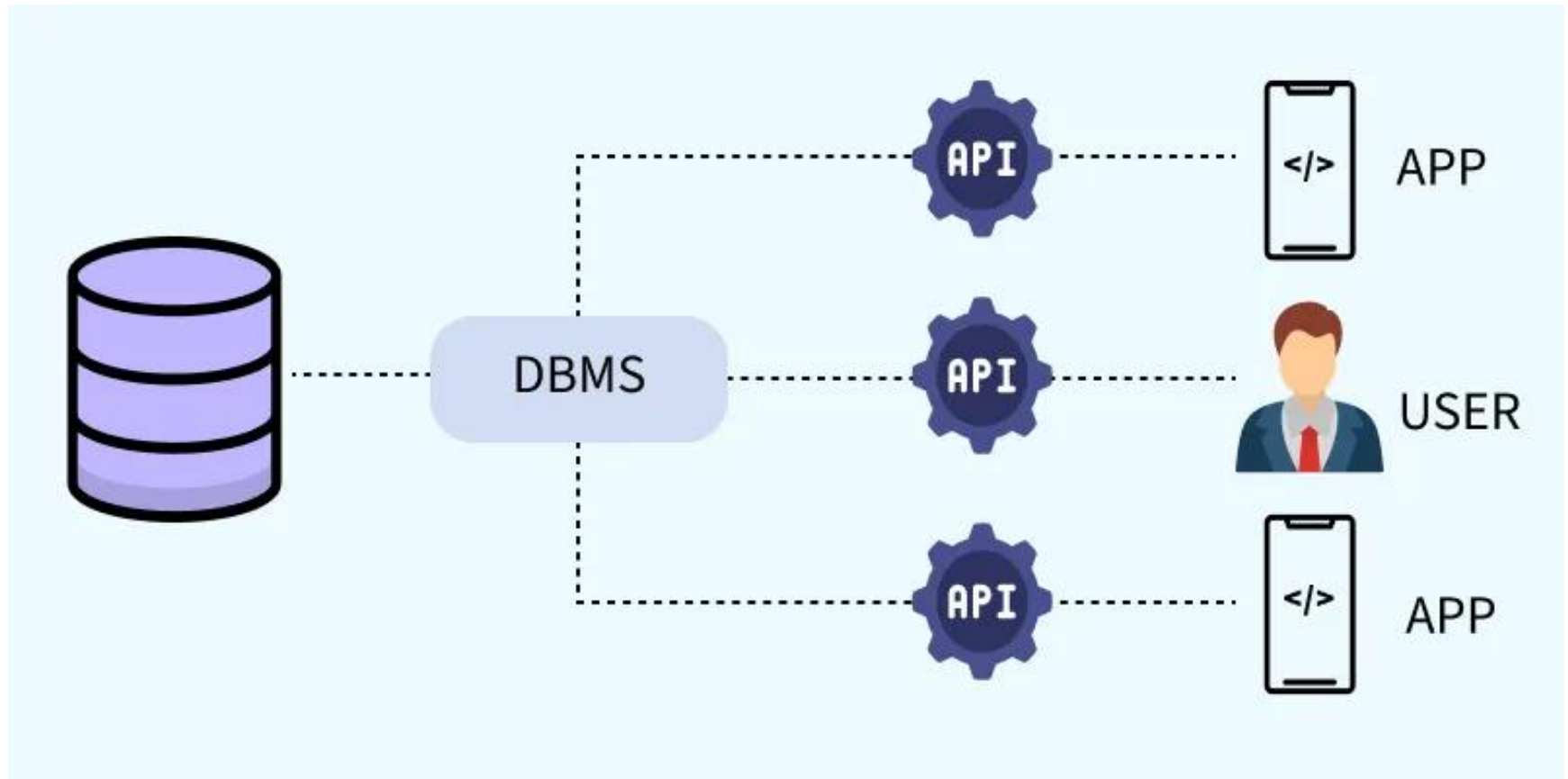
Disadvantages of file processing

- **Data isolation:** Because data are scattered in various files, and files may be in different formats, writing new application programs to retrieve the appropriate data is difficult.
- **Dependency** on application programs: Changing files would lead to change in application programs.
- **Limited** time sharing
- **Lengthy** development times
- Excessive program maintained.

Disadvantages of file processing



Solution: the database approach



Example of a simplified database catalog

RELATIONS

Relation_name	No_of_columns
STUDENT	4
COURSE	4
SECTION	5
GRADE_REPORT	3
PREREQUISITE	2

Figure 1.3

An example of a database catalog for the database in Figure 1.2.

COLUMNS

Column_name	Data_type	Belongs_to_relation
Name	Character (30)	STUDENT
Student_number	Character (4)	STUDENT
Class	Integer (1)	STUDENT
Major	Major_type	STUDENT
Course_name	Character (10)	COURSE
Course_number	XXXXNNNN	COURSE
....		
....		
....		
Prerequisite_number	XXXXNNNN	PREREQUISITE

Note: Major_type is defined as an enumerated type with all known majors. XXXXNNNN is used to define a type with four alpha characters followed by four digits

Main Characteristics of the Database Approach (continued)

- **Support of multiple views of the data:**
 - Each user may see a different view of the database, which describes **only** the data of interest to that user.

Main Characteristics of the Database Approach (continued)

- **Sharing of data and multi-user transaction processing:**
 - Allowing a set of **concurrent users** to retrieve from and to update the database.
 - *Concurrency control* within the DBMS guarantees that each **transaction** is correctly executed or aborted
 - *Recovery* subsystem ensures each completed transaction has its effect permanently recorded in the database
 - **OLTP** (Online Transaction Processing) is a major part of database applications. This allows hundreds of concurrent transactions to execute per second.

Database Users

- Users may be divided into
 - Actors on the Scene: These are the people who use the database and work with its data.
 - They also create, update, and maintain database applications.
 - Workers Behind the Scene: These are the people who build and maintain the DBMS itself and operate and manage the computer systems.

Database Users

- Actors on the scene
 - **Database administrators:**
 - Responsible for authorizing access to the database, for coordinating and monitoring its use, getting software and hardware resources, controlling its use and monitoring efficiency of operations.
 - **Database Designers:**
 - Responsible to define the content, the structure, the constraints, and functions or transactions against the database. They must communicate with the end-users and understand their needs.

Categories of End-users

- Actors on the scene (continued)
 - **End-users:** They use the data for queries, reports and some of them update the database content. End-users can be categorized into:
 - **Casual:** They use the database only sometimes, when they need specific information
 - **Naïve** They are a large group of users who use the database through predefined operations. They do not write queries themselves; they just follow fixed steps.
 - Examples are bank-tellers or reservation assistants who do this activity for an entire shift of operations.

Summary

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